UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	INTRODUCTION TO INTERACTIVE MULTIMEDIA
COURSE CODE	CSC413
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

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This examination paper consists of 5 printed pages.

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PART A (15 MARKS)

State TRUE (T) or FALSE (F) for the following statements.

1. A vector image becomes pixelated when enlarged.
2. Interlacing displays every line of a video frame at the same time.
3. Multimedia combines text, graphics, audio, video and animation.
4. Anti-aliasing makes the edges of text look more jagged.
5. A plotter is an input device.
6. RAM is a type of permanent storage.
7. A linear presentation allows users to choose their own path.
8. Path animation moves an object along a predefined route.
9. Interactivity level is irrelevant when choosing an authoring tool.
10. Frame rate is the number of video frames displayed per second.
11. A scanner is an input device.
12. RGB is an additive colour model.
13. Tweening creates intermediate frames between keyframes.
14. CMYK is used for printed output.
15. A URL identifies the address of a web resource.

PART B (15 MARKS)

Select the most appropriate answer from Table A for each of the following questions.

Table A

A. Multimedia	B. Hypermedia	C. Keyframe	D. Prototype	E. Copyright
F. Card-based tool	G. Storyboard	H. MIDI	I. Sampling	J. Bitmap
K. Linear multimedia	L. Browser	M. Tweening	N. Cel animation	O. Scanner

1. visual sequence plan
2. important frame in an animation
3. integration of text, graphics, audio, video and animation
4. generating in-between frames
5. exclusive rights of a creator
6. presentation without user control of sequence
7. hand-drawn frame-by-frame animation
8. input device that digitises documents
9. authoring tool organised like pages or cards
10. converting sound into digital values
11. image made of pixels
12. early model of a product
13. software that displays web pages
14. standard for musical notes and instructions
15. linked multimedia elements

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PART C (30 MARKS)

Choose the correct answer for each question.

1. MIDI stores:
A. Images
B. Video frames
C. Musical notes and instructions
D. Digitized audio samples

2. Sampling converts:
A. Images to vectors
B. Digital video to text
C. Text to speech
D. Analog sound to digital values

3. Which is a 3D animation task?
A. Modelling and rendering
B. Bookmarking
C. Kerning
D. Sampling

4. HTML is used to:
A. Store audio samples
B. Compress video
C. Model 3D objects
D. Structure web content

5. CMYK is used mainly for:
A. Animation
B. Printed output
C. Sound mixing
D. Screen display

6. A bitmap image is made of:
A. Sound samples
B. Mathematical objects
C. Text characters
D. Pixels

7. Interlacing:
A. Draws alternate lines of a frame
B. Adds sound
C. Removes colour
D. Draws every line at once

8. Which is an authoring tool category?
A. CMYK
B. RGB
C. Card-based
D. JPEG

9. Which is an audio file format?
A. MPEG
B. HTML
C. WAV
D. JPEG

10. An authoring tool is used to:
A. Assemble multimedia elements
B. Only record audio
C. Only print text
D. Only browse the web

11. Kerning adjusts:
A. Page size
B. Font colour
C. Space between individual character pairs
D. Line length

12. Which is NOT a multimedia element?
A. Text
B. Audio
C. Compiler
D. Animation

13. A serif font has:
A. Only uppercase letters
B. No spacing
C. No strokes
D. Small strokes at character ends

14. Which is a benefit of multimedia in education?
A. It combines visuals and sound
B. It removes all interaction
C. It uses only text
D. It replaces all teachers

15. Chroma key is used to:
A. Add text
B. Scan images
C. Compress audio
D. Replace a colour background

16. A storyboard is:
A. A visual sequence plan
B. A sound file
C. A colour model
D. A video codec

17. Which best defines multimedia?
A. Text only
B. A printed book
C. A single image
D. Text, graphics, audio, video and animation

18. Which image type scales without losing quality?
A. Scan
B. Vector
C. Bitmap
D. Photograph

19. A higher sampling rate generally:
A. Reduces accuracy
B. Improves accuracy
C. Changes colour
D. Removes audio

20. Cel animation is created by:
A. Printing images
B. Scanning text
C. Drawing each frame by hand
D. Recording sound

21. Hypermedia is best described as:
A. Text-only media
B. A single video file
C. A printed manual
D. Multimedia with hyperlinks

22. RAM is:
A. Temporary working memory
B. An input device
C. A web browser
D. Permanent storage

23. Virtual reality creates:
A. A printed page
B. A text font
C. A simulated environment
D. A camera lens

24. Resolution refers to:
A. Font size
B. File type
C. Number of pixels in an image
D. Colour name

25. Which is an example of linear multimedia?
A. A cinema presentation
B. A game with choices
C. An interactive kiosk
D. A website with menus

26. A browser is used to:
A. Scan images
B. Edit audio
C. Print documents only
D. Display web pages

27. A hyperlink links to:
A. Only images
B. Only audio
C. Another resource
D. Only text

28. A graphics tablet is used for:
A. Sound output
B. Web browsing
C. Printing
D. Drawing input

29. MPEG is:
A. An image editor
B. A storage device
C. A video compression family
D. A font

30. A wireframe is:
A. A codec
B. A final video
C. A low-fidelity layout sketch
D. A sound file

PART D (40 MARKS)

QUESTION 1

Text is an essential element in a multimedia project. Its shape, size, design, colour, arrangement and linking all affect the message.

a) State TWO reasons sans serif is mostly used for screen display and serif is used for printed material.

(2 marks)

b) Describe TWO roles of hypertext in multimedia.

(3 marks)

c) Explain how text can have a massive impact on an informative multimedia project in the link and

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HTML context.

(5 marks)

QUESTION 2

A shopping complex requires a kiosk that gives customers location and retailer information.

a) State TWO important project team roles for completing the kiosk.

(3 marks)

b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

(3 marks)

c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

(4 marks)

QUESTION 3

A product catalogue must show sharp images on screen and in print.

a) State TWO differences between bitmap and vector images.

(3 marks)

b) Describe TWO factors that affect image quality.

(3 marks)

c) Explain how colour model and resolution affect screen and print output.

(4 marks)

QUESTION 4

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

(2 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

(4 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

(4 marks)

END OF QUESTION PAPER